Conditional Probability KEY

Name _____

A **conditional probability** is a probability when you already know something happened.

$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ read "the probability of A **given** B"

In a table: only look at the B group. $P(A \mid B) = \dfrac{\text{number in } A \text{ and } B}{\text{number in } B}$

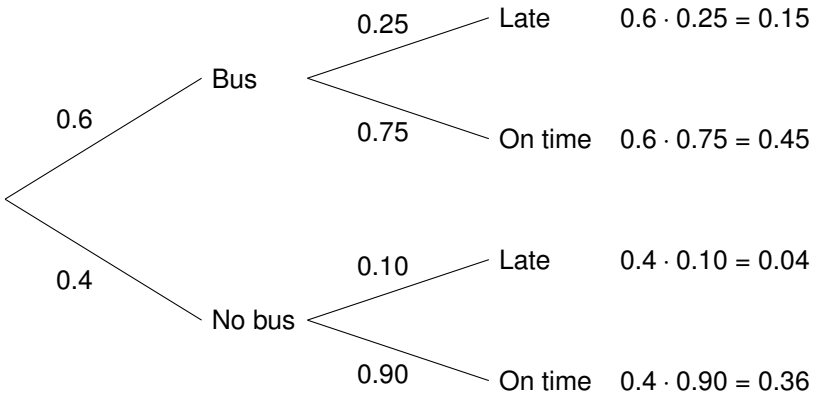
	Job	**No job**	**Total**
Sport	8	22	30
No sport	12	18	30
Total	20	40	60

$P(\text{job} \mid \text{sport}) = \dfrac{8}{30} \approx 0.27$ (only the 30 in sports)

$P(\text{sport} \mid \text{job}) = \dfrac{8}{20} = 0.40$ (only the 20 with jobs)

The order matters. These are different questions.

Tree diagrams. 60% of students ride the bus. 25% of bus riders are late. 10% of the others are late.



Multiply **along** a path: $P(\text{bus} \cap \text{late}) = 0.6 \cdot 0.25 = 0.15$

Add the paths you want: $P(\text{late}) = 0.15 + 0.04 = 0.19$

Then divide: $P(\text{bus} \mid \text{late}) = \dfrac{0.15}{0.19} \approx 0.79$

About 79% of late students ride the bus. Only 60% of all students ride the bus. So knowing a student is late makes it more likely that the student rides the bus.

Independence check. A and B are independent if $P(A \mid B) = P(A)$.

$P(\text{job}) = \dfrac{20}{60} \approx 0.33$ but $P(\text{job} \mid \text{sport}) \approx 0.27$.

They are different, so having a job and playing a sport are **not** independent.

Work with a partner. Compare answers when you finish each problem.

1. 100 students said if they ate breakfast and if they were on time.

	Breakfast	**No breakfast**	**Total**
On time	45	15	60
Late	10	30	40
Total	55	45	100

   (a) $P(\text{late})$ $\frac{40}{100} = 0.40$

   (b) $P(\text{late} \mid \text{no breakfast})$ $\frac{30}{45} \approx 0.67$

   (c) $P(\text{no breakfast} \mid \text{late})$ $\frac{30}{40} = 0.75$

   (d) $P(\text{late} \mid \text{breakfast})$ $\frac{10}{55} \approx 0.18$

   (e) Are being late and eating breakfast independent? Explain.
   No. $P(\text{late}) = 0.40$, but $P(\text{late} \mid \text{no breakfast}) \approx 0.67$. Knowing about breakfast changes the probability.

2. $P(A \cap B) = 0.12$ and $P(B) = 0.4$. Find $P(A \mid B)$.
   $\frac{0.12}{0.4} = 0.3$

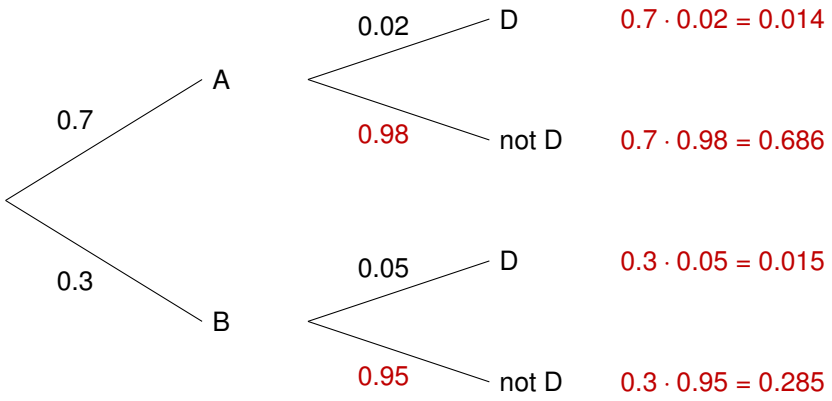
3. Draw one card from a deck. Face cards are J, Q, and K (12 in all).

   (a) $P(\text{king} \mid \text{face card})$ $\frac{4}{12} = \frac{1}{3}$

   (b) $P(\text{face card} \mid \text{king})$ $\frac{4}{4} = 1$. Every king is a face card.

4. A factory has two machines. Machine A makes 70% of the parts and Machine B makes 30%. 2% of A's parts are defective (D). 5% of B's parts are defective.

   (a) Fill in the tree diagram.



   (b) Find $P(D)$, the probability a part is defective. $0.014 + 0.015 = 0.029$

   (c) A part is defective. Find $P(B \mid D)$.
   $\frac{0.015}{0.029} \approx 0.52$. Even though B makes fewer parts, about half of the defective parts come from B.

5. 100 students said if they wear glasses and if they play an instrument. 40 play an instrument, and 12 of them wear glasses. 30 of all 100 students wear glasses. Are wearing glasses and playing an instrument independent? Show your work.

   $P(\text{glasses}) = \frac{30}{100} = 0.3$. $P(\text{glasses} \mid \text{instrument}) = \frac{12}{40} = 0.3$. They are equal, so yes, independent.